



PRISM

Plasma **R**esearch **I**ntegrated **S**ystem for **M**ulti-diagnostics

v2.1.0

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Overview

What is PRISM

PRISM is a software platform for viewing and analyzing multi-diagnostic data from the **KSTAR tokamak** in a [unified interface](#).

It directly connects to the MDS+ database to load experimental data, providing various visualizations from profile analysis to spectrograms and imaging analysis.

How to Run: Execute `$ prism` command on nkstar server

Key Numbers

11

Diagnostics

11

Tabs

5

EFIT Trees

2

Themes (Dark/Light)

Supported Diagnostics

DIAGNOSTIC	FULL NAME	CHANNELS	PARAMETERS	TABS
CES	Charge Exchange Spectroscopy	32	T_i, v_T	Profile Time Trace
XICS	X-ray Imaging Crystal Spectrometer	1	T_{i0}, v_{T0}	Profile Time Trace
TS	Thomson Scattering	31	T_e, n_e	Profile Time Trace
ECE	Electron Cyclotron Emission	76	T_e	Profile Time Trace Spectrogram
MSE	Motional Stark Effect	25	γ, q, j	Profile Time Trace
BES	Beam Emission Spectroscopy	4x16	Intensity	Spectrogram
TCI	Two-Color Interferometer	5	n_e (line)	Spectrogram Time Trace
ECEI	ECE Imaging	24x8x3	T_e fluct.	Spectrogram
MC	Mirnov Coil	20T/22P	dB/dt	Spectrogram
TV	In-Vessel Inspection System TV	-	Video	Imaging
IRVB	Infra-Red Video Bolometer	-	P_{rad}	Imaging

Key Features



11 Diagnostics

CES, TS, ECE, MSE, BES, TCI, ECEI, MC, TV, IRVB integrated in one UI



Dark/Light Themes

Toggle via bottom toolbar for all widgets and plots



EFIT Mapping

5 EFIT trees for R → flux coordinate transformation



Data Export

Preview spreadsheet than save as .csv or .npz or .png



Plot Options

Colormap, font sizes per tab.
Settings persist across sessions



Channel Toggle

Show/high individual channels in Profile to highlight specific data



On-demand Loading

Tabs initialize only when selected for fast startup



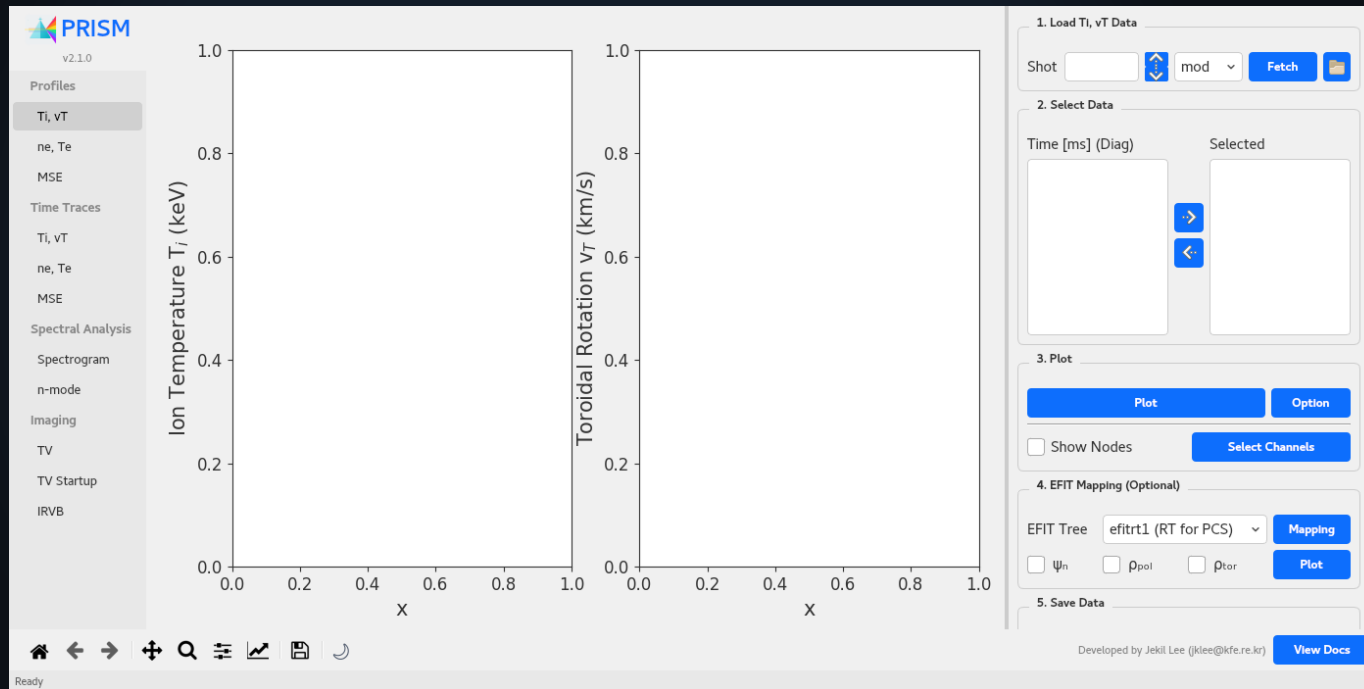
What's New

Auto-popup on update shows changelog

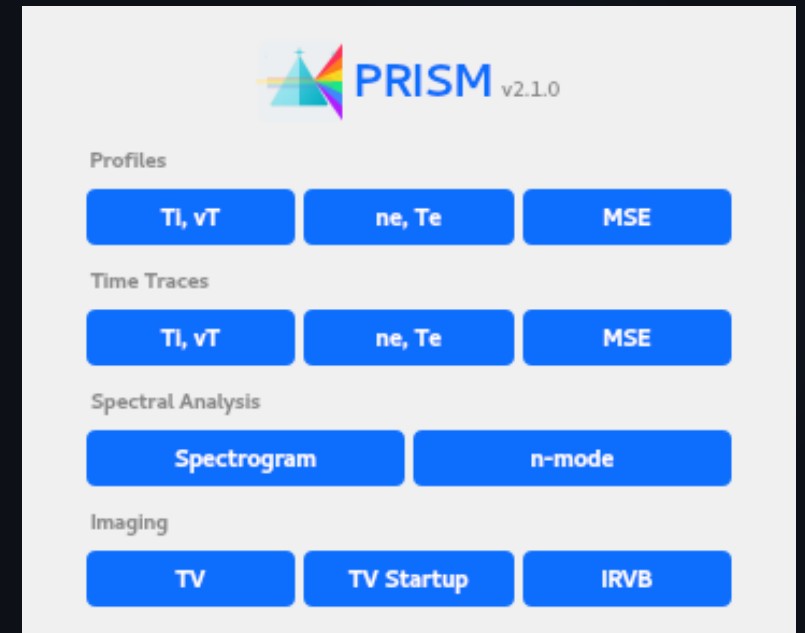
User Interface

UI Structure

Full Mode `$ prism`



Selector Mode `$ prism -s`



● **Sidebar** 4 categorizes, 11 tabs

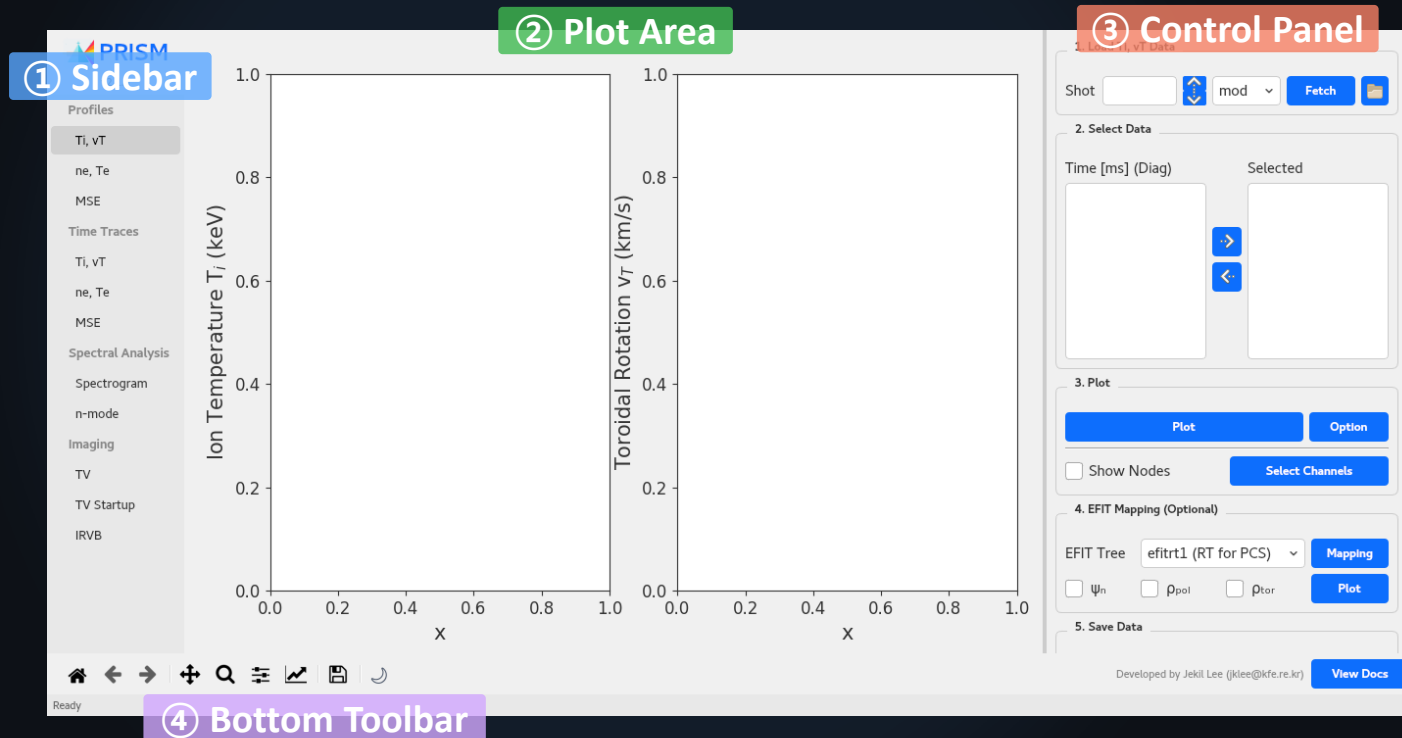
● **Control Panel** Shot/Fetch/Select/Plot

● **Plot Area** matplotlib interactive

● **Bottom Toolbar** Pan/Zoom, axis range, Docs

User Interface

GUI Function Guide



① Sidebar

- **Profile:** T_i , v_T / n_e , T_e / MSE profiles
- **Time Traces:** Time-series plots
- **Spectral Analysis:** Spectrogram, n-mode
- **Imaging:** TV, Startup, IRVB

③ Control Panel

1. **Load Data:** Enter shot number, Fetch button
2. **Select Data:** Time/channel selection
3. **Plot:** Execute plot, Options (font size, etc...)
4. **EFIT Mapping:** Flux coordinate transform
5. **Save Data:** CSV export

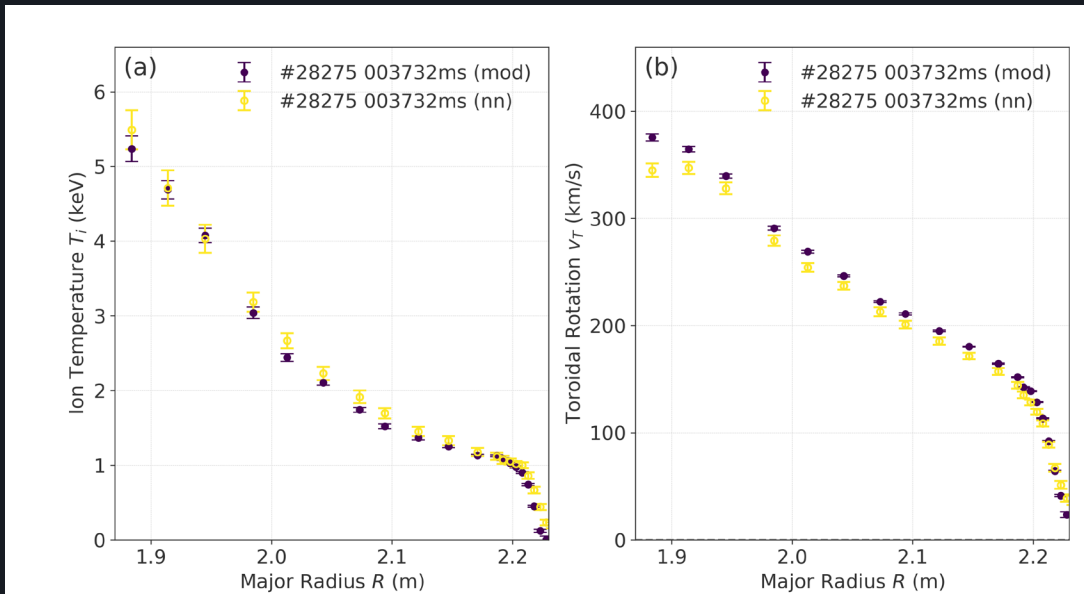
④ Bottom Toolbar

- | | |
|-----------------------------|-------------------------------------|
| Home: Reset view | Back/Forward: Navigate views |
| Pan: Drag to move | Zoom: Zoom to region |
| Axes: Set axis range | Margin: Set margin range |
| Save: Save image | Theme: Dark/Light |

Profiles

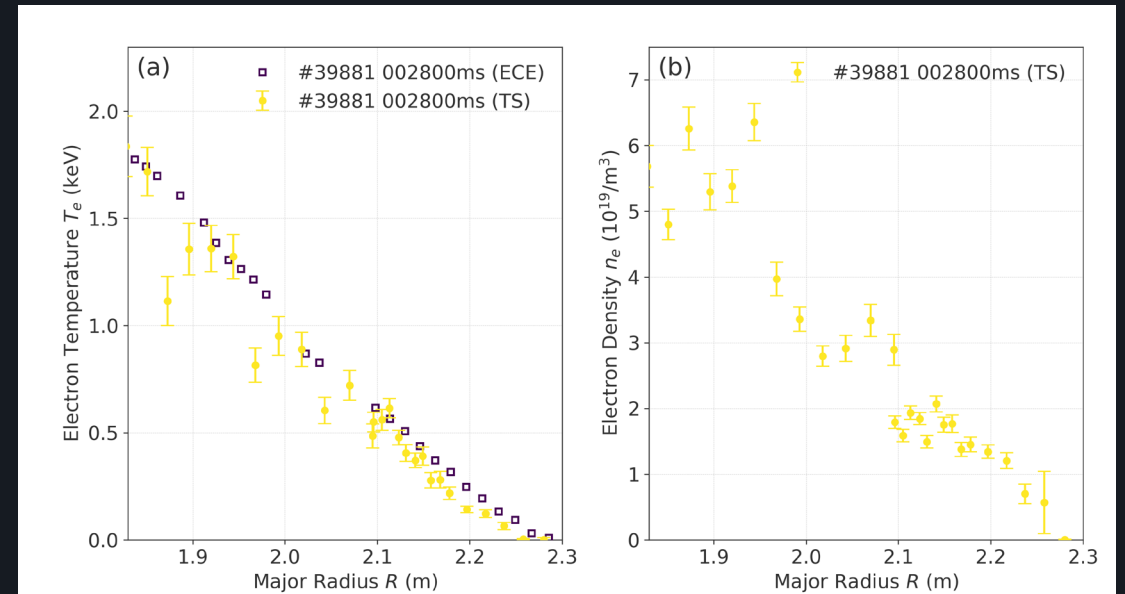
T_i , v_T Profile & n_e , T_e Profile

Profile T_i , v_T Ion Temperature & Toroidal Rotation



- **CES (*mod/nn) + XICS overlay**
* mod: Beam Modulation / nn: Neural Network [[J. K. Lee FED \(2026\)](#)]

Profile n_e , T_e Electron Density & Temperature



- **TS + ECE overlay**
- **Auto channel position**

Common Features EFIT coordinate mapping (5 trees) · 9 colormaps · Font size settings · Data preview/export (.csv, .npz)

MSE Profile

Measured Parameter

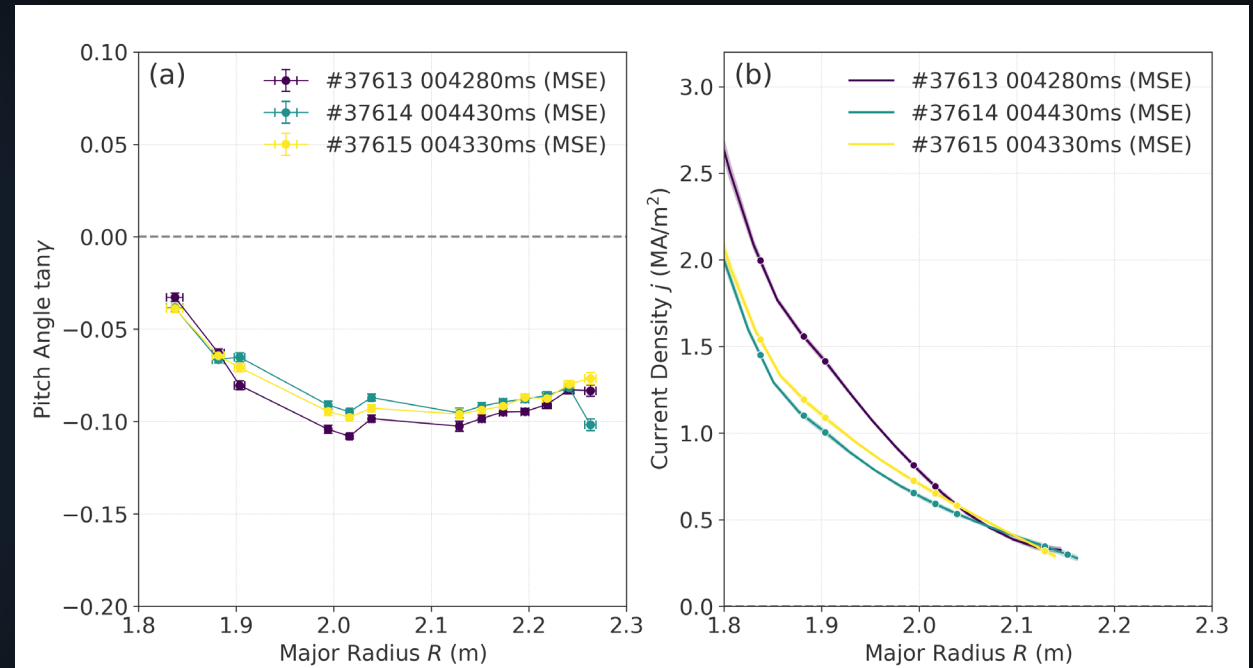
- **Pitch Angle ($\tan \gamma$):** MSE raw data (25 channels)

Derived Parameters [[C. C. Petty NF \(2002\)](#)]

- **q Profile:** Safety factor
- **j Profile:** Current density

Special Features

- Auto NB source detection (NB-1A / 1B / 1C)



Time Traces

Time Trace Tabs

T_i , v_T Time Trace

CES / XICS

- CES channel-wise T_i , v_T evolution
- XICS overlay available

n_e , T_e Time Trace

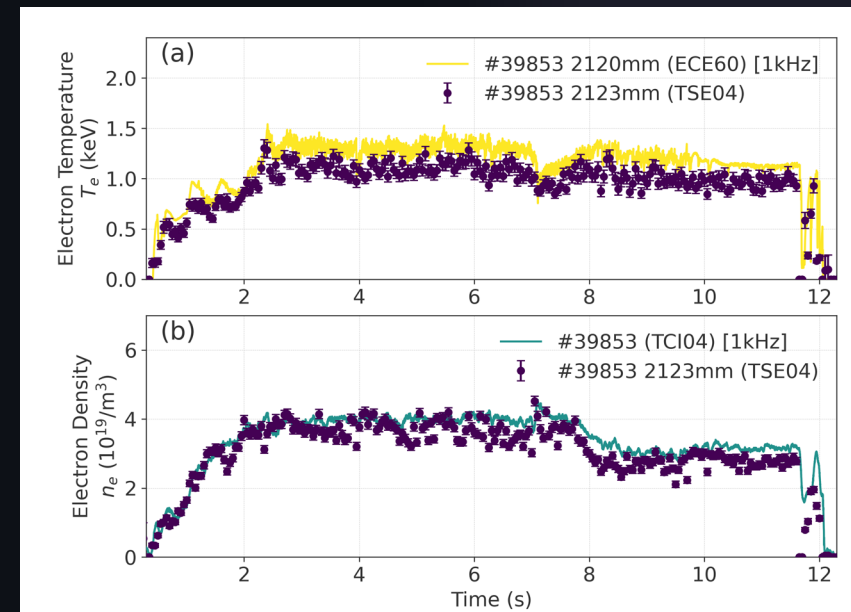
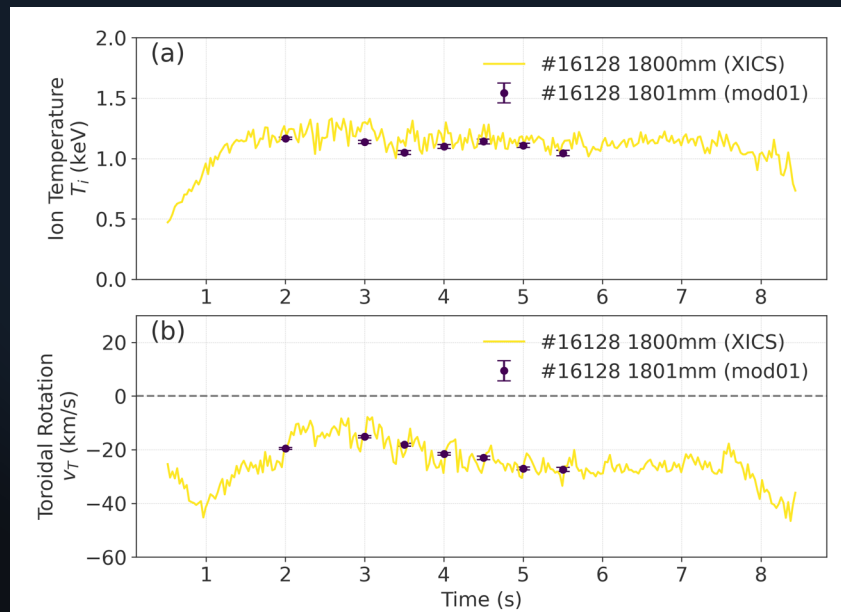
Thomson / ECE / TCI

- TS/ECE channel-wise T_e , n_e
- Diagnostic selection available

MSE Time Trace

MSE

- $\tan \gamma$, q , j time evolution
- q/j toggle radio buttons



Spectrogram

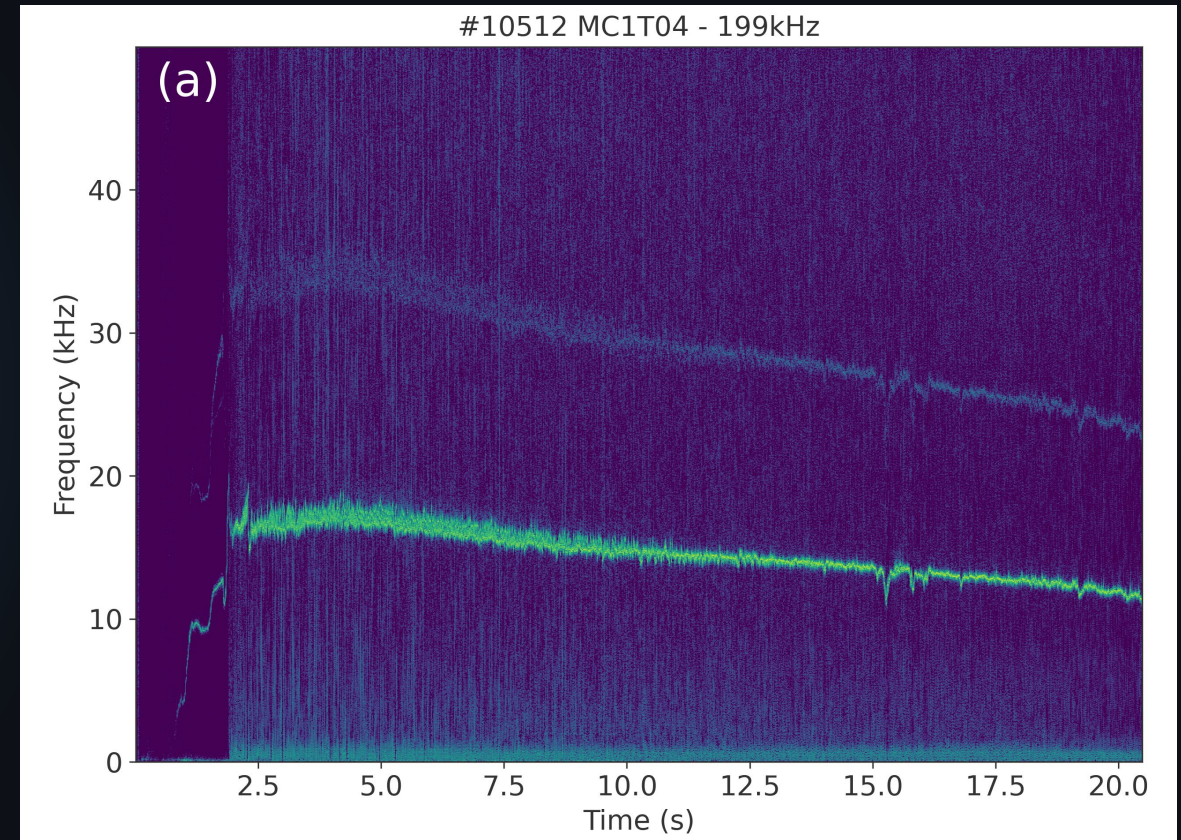
Spectrogram

Supported Diagnostics

ECE	Electron Cyclotron Emission	76 ch
MSE	Motional Stark Effect	25
BES	Beam Emission Spectroscopy	4x16 ch
TCI	Two-Color Interferometer	5 ch
ECEI	ECE Imaging	24x8x3 ch
MC	Mirnov Coil	20T/22P ch

Configuration Parameters

- NFFT, dynamic range settings
- Time range & frequency range settings
- NPZ data export



Spectrogram

n-Mode Spectrum

Data Source Measured Parameter

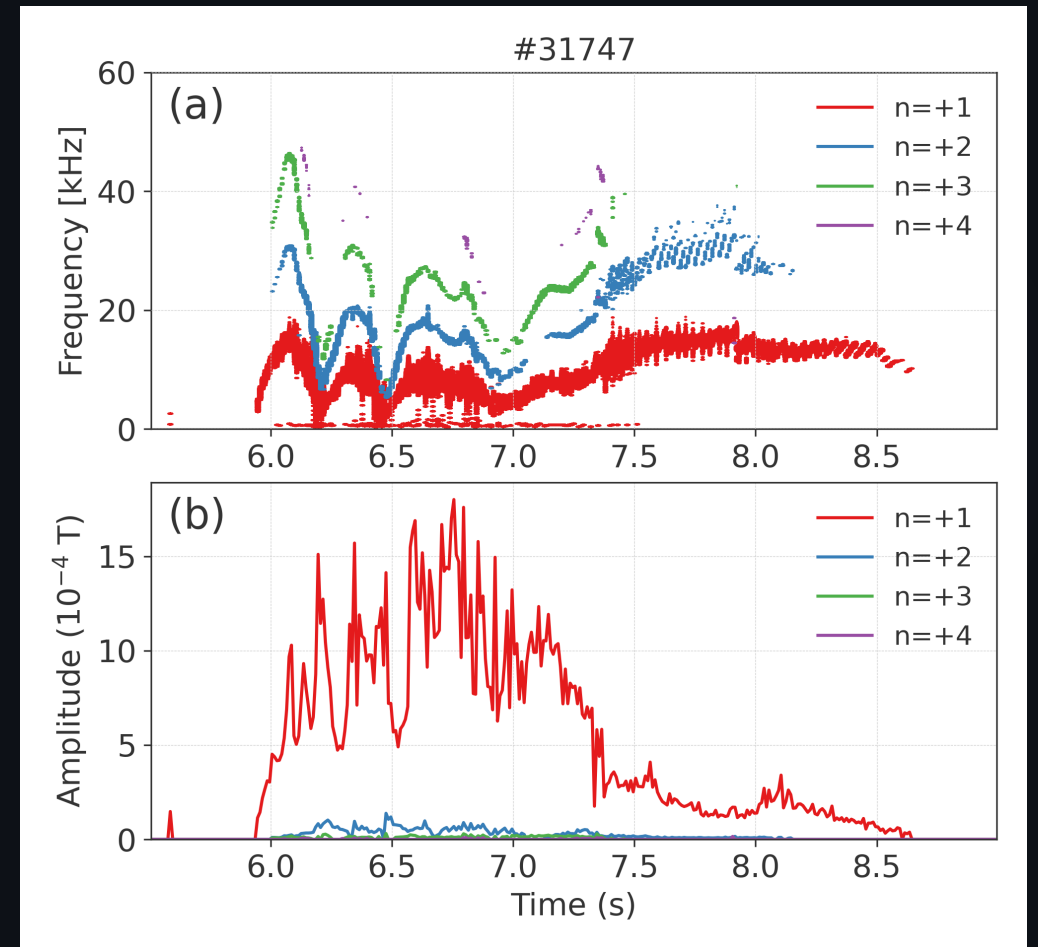
FFT analysis of dB/dt data from MC1T (toroidal Mirnov coils) to calculate amplitude spectrum for toroidal mode numbers

Analysis Options

- Mode Sign, Integrate, Detrend, Plot Type, Color

Output

- Top: Time-frequency map
- Bottom: Mode amplitude
- NPZ export supported



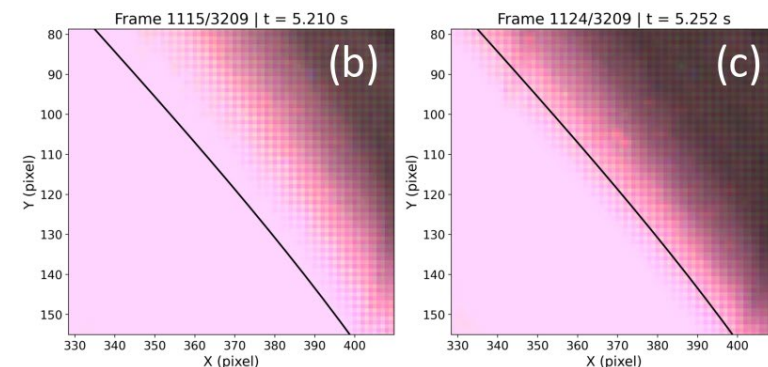
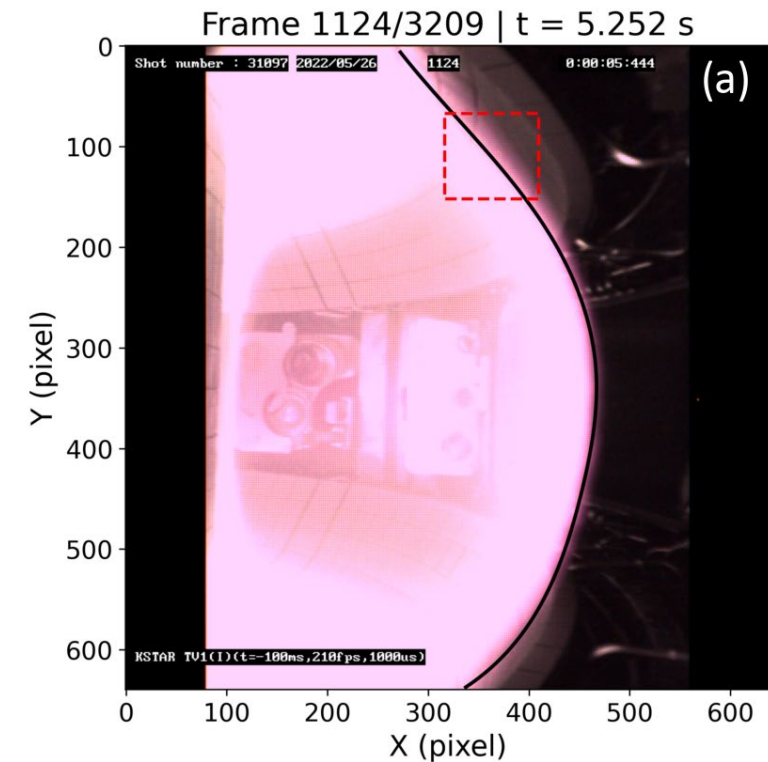
Imaging

TV

Key Features

- Auto-load images by shot number
- Frame slider / prev-next / direct input / mouse wheel
- Auto-display time (seconds) for each frame
- Compare mode: TV01 + TV02 side-by-side
- Play/Pause/Stop playback control, FPS adjustment
- Draw Mode: Spline interpolation curve drawing

* Related study: Edge kink-like mode observation [[J. K. Lee PoP \(2025\)](#)]

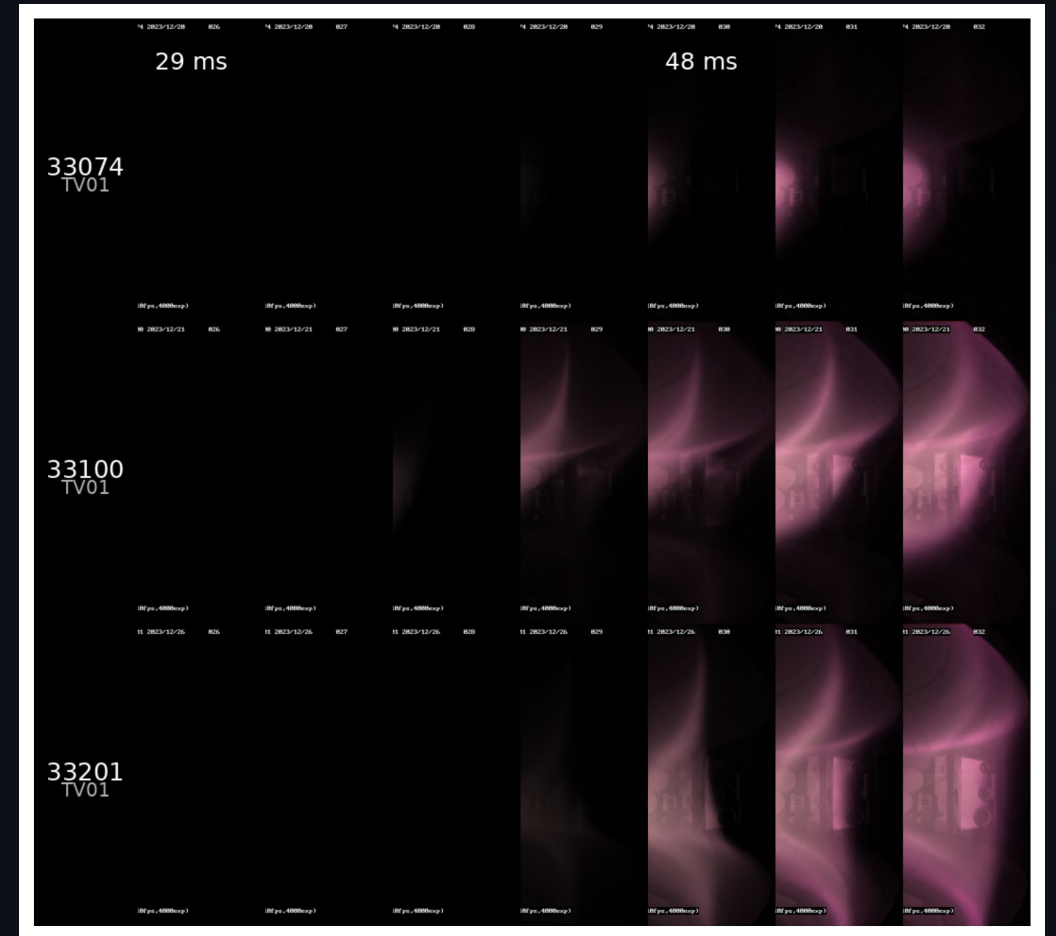


TV Startup

Key Features

- Compare TV images from multiple shots in **grid layout**
- TV01/TV02 **free mixing** comparison
- Set frame range or time range

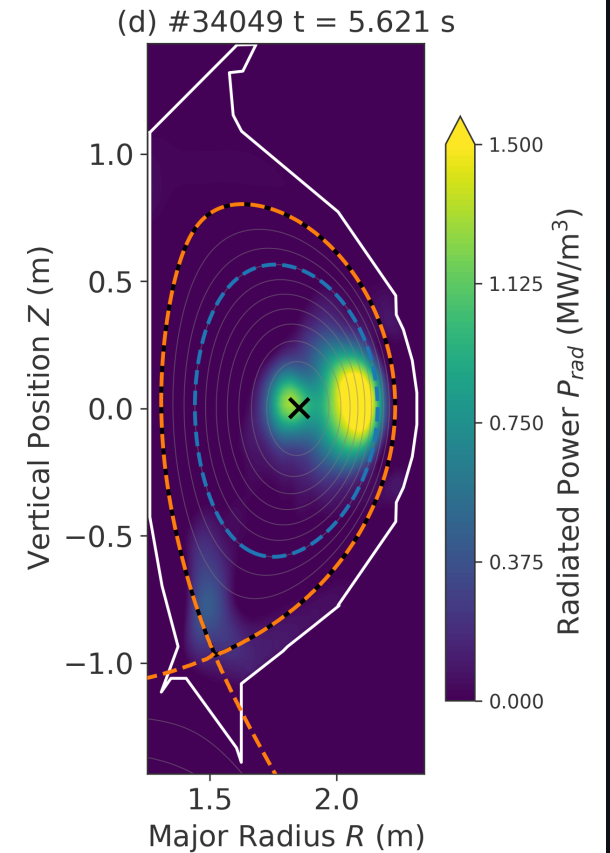
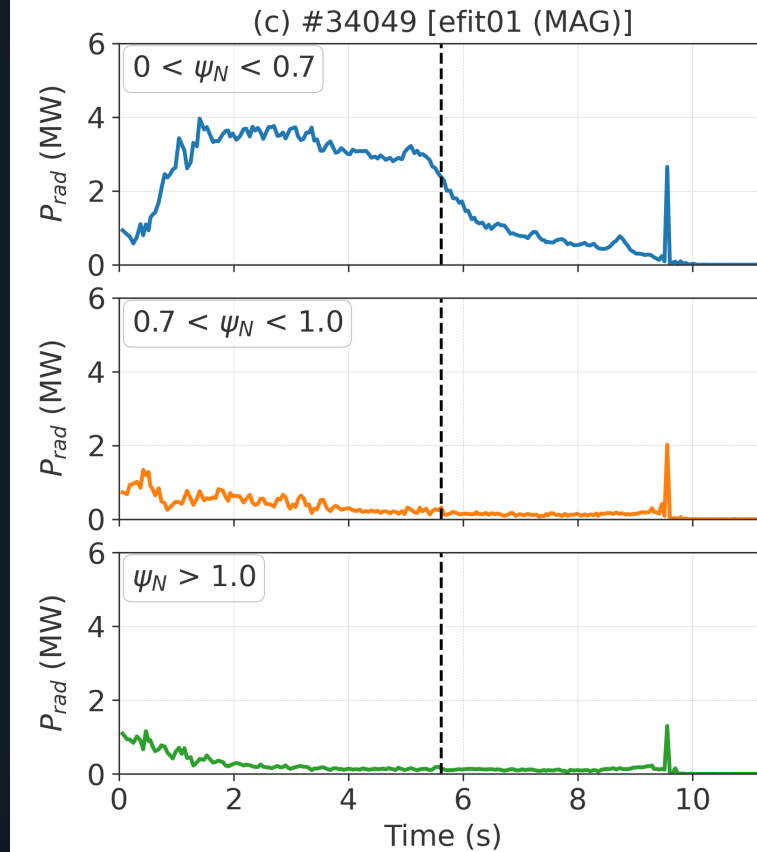
* Related study: Startup Scenario Optimization [[J. Lee FED \(2024\)](#)]



IRVB

Key Features

- 2D P_{rad} heatmap
- EFIT overlay
- User-defined flux region analysis
- Region-wise P_{rad} traces
- Frame navigation



Common Features

Data Export & EFIT Mapping



Data Export

Preview plotted data in spreadsheet format before saving.

.csv Format

Tab-delimited CSV file. Open directly in spreadsheet

.npz Format

NumPy compressed binary + example script auto-generated

Data Copy

Copy data to clipboard from preview into Excel, etc.



EFIT Coordinate Mapping

Transform x-axis from real-space to flux coordinates.

Coordinate Types

ψ_N

ρ_t

ρ_p

EFIT Trees (5 types)

efitrt1

Magnetic constraints for real-time

efitrt2

Magnetic and MSE constraints for real-time

efit01

Magnetic constraints

efit02

Magnetic and MSE constraints

efit04

Drift-corrected magnetic constraints



Theme

Dark/Light themes available



Plot Options

Colormaps, font sizes



What's New

Auto-popup changelog after updates



Thank You!

PRISM

Plasma Research Integrated System for Multi-diagnostics

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